

Neural-Symbolic Reasoning over Knowledge Graphs: A Survey from a Query Perspective

Lihui Liu
lihui2@illinois.edu
Wayne State University
Detroit, Michigan, USA

Zihao Wang*
zihao@illinois.edu
University of Illinois at Urbana
Champaign
Urbana, Illinois, USA

Hanghang Tong
htong@illinois.edu
University of Illinois at Urbana
Champaign
Urbana, Illinois, USA

ABSTRACT

Knowledge graph reasoning is pivotal in various domains such as data mining, artificial intelligence, the Web, and social sciences. These knowledge graphs function as comprehensive repositories of human knowledge, facilitating the inference of new information. Traditional symbolic reasoning, despite its strengths, struggles with the challenges posed by incomplete and noisy data within these graphs. In contrast, the rise of Neural Symbolic AI marks a significant advancement, merging the robustness of deep learning with the precision of symbolic reasoning. This integration aims to develop AI systems that are not only highly interpretable and explainable but also versatile, effectively bridging the gap between symbolic and neural methodologies. Additionally, the advent of large language models (LLMs) has opened new frontiers in knowledge graph reasoning, enabling the extraction and synthesis of knowledge in unprecedented ways. This survey offers a thorough review of knowledge graph reasoning, focusing on various query types and the classification of neural symbolic reasoning. Furthermore, it explores the innovative integration of knowledge graph reasoning with large language models, highlighting the potential for groundbreaking advancements. This comprehensive overview is designed to support researchers and practitioners across multiple fields, including data mining, AI, the Web, and social sciences, by providing a detailed understanding of the current landscape and future directions in knowledge graph reasoning.

CCS CONCEPTS

• **Computing methodologies** → **Reasoning about belief and knowledge**; • **Information systems** → **Data mining**.

KEYWORDS

Knowledge graph reasoning

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*ZW contributed during his visit to UIUC in 2023-2024

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1 INTRODUCTION

A knowledge graph is a graph structure that contains a collection of facts, where nodes represent real-world entities, events, and objects, and edges denote the relationships between two nodes. It is a powerful tool for organizing and connecting information in a way that mimics human thought and learning. Since its debut in 2012,¹ a variety of knowledge graphs have been generated, including Freebase, Yago, and Wikidata.

Knowledge graph reasoning refers to the process of deriving new knowledge or insights from existing knowledge graphs in response to a query. In essence, the knowledge graph reasoning pipeline comprises three key elements: the input query, background knowledge, and reasoning model, each posing unique challenges. The background knowledge may vary in completeness, influencing the system's ability to accurately interpret and utilize information. Meanwhile, input queries range from clear and specific to ambiguous or dynamically changing, demanding robust mechanisms for understanding user intent. Furthermore, the reasoning model's approach, whether inductive or deductive, impacts the system's ability to draw meaningful conclusions from the available data. Addressing these challenges necessitates adaptable algorithms and techniques to ensure the efficacy and reliability of knowledge graph reasoning across diverse contexts and applications.

Recently, there is a trend to utilize neural-symbolic artificial intelligence to enhance reasoning on knowledge graphs. Since knowledge graphs can be viewed as discrete symbolic representations of knowledge, it is natural to integrate knowledge graphs with neural models to unleash the full potential of neural-symbolic reasoning. Traditional symbolic reasoning is intolerant of ambiguous and noisy data, but knowledge graphs are often incomplete, which brings difficulties to symbolic reasoning. On the contrary, the recent advances in deep learning promote neural reasoning on knowledge graphs, which is robust to ambiguous and noisy data but lacks interpretability compared to symbolic reasoning. Considering the advantages and disadvantages of both methodologies, recent efforts have been made to combine the two reasoning methods for better reasoning on knowledge graphs.

Last but not least, the emergence of large language models (LLMs) has shown impressive natural language capabilities. However, they struggle with logical reasoning that requires structured knowledge. The integration of LLMs with knowledge graphs during the reasoning process represents a promising area of exploration. While some methods have been proposed in this regard, a large part of this topic is unexplored or under-explored.

¹https://en.wikipedia.org/wiki/Knowledge_graph

This survey provides a comprehensive exploration of knowledge graph reasoning for different types of queries. We introduce knowledge graph reasoning for single-hop queries, complex logical queries, and natural language queries. Furthermore, the integration of neural symbolic artificial intelligence techniques is investigated, highlighting innovative methodologies for enhancing reasoning capabilities within knowledge graphs. Lastly, the survey delves into the fusion of Large Language Models (LLMs) with knowledge graph reasoning, showcasing advancements at the intersection of natural language processing and structured data reasoning.

While numerous surveys have explored knowledge graph embedding, few have explicitly addressed knowledge graph reasoning from both query types and neural symbolic perspectives. This paper aims to fill this gap by introducing different topics and offering a comprehensive overview of knowledge graph reasoning from diverse viewpoints. Through detailed classification and elaboration within each category, this paper provides a holistic understanding of the complexities and advancements in knowledge graph reasoning, bridging the gap between different query types and neural symbolic reasoning, and offering insights into future directions for this field.

The remainder of the article is structured as follows. Section 2 provides an overview of the background knowledge closely associated with knowledge graph reasoning and neural symbolic reasoning. Section 3 delves into knowledge graph reasoning for single-hop queries, examining various perspectives. Following this, Section 4 explores the intricacies of reasoning with complex logical queries. In Section 5, we scrutinize knowledge graph reasoning for both single-turn and multi-turn queries in natural language. Finally, we draw conclusions based on the insights gained from our survey.

2 PRELIMINARY

In this section, we first formally define knowledge graphs and knowledge graph reasoning.

2.1 brief history of knowledge graph

Knowledge representation has experienced a long-period history of development in the fields of logic and AI. The idea of graphical knowledge representation firstly dated back to 1956 as the concept of semantic net proposed by Richens [10], while the symbolic logic knowledge can go back to the General Problem Solver [1] in 1959. The knowledge base is firstly used with knowledge-based systems for reasoning and problemsolving. MYCIN [2] is one of the most famous rule-based expert systems for medical diagnosis with a knowledge base of about 600 rules. Later, the community of human knowledge representation saw the development of frame-based language, rule-based, and hybrid representations. Approximately at the end of this period, the Cyc project¹ began, aiming at assembling human knowledge. Resource description framework (RDF)² and Web Ontology Language (OWL)³ were released in turn, and became important standards of the Semantic Web⁴. Then, many open knowledge bases or ontologies were published, such as WordNet, DBpedia, YAGO, and Freebase. Stokman and Vries [7] proposed a modern idea of structure knowledge in a graph in 1988. However, it was in 2012 that the concept of knowledge graph gained great popularity since its first launch by Google's search engine⁵, where

the knowledge fusion framework called Knowledge Vault [3] was proposed to build large-scale knowledge graphs. A brief road map of knowledge base history is illustrated in Fig. 10 in Appendix A. Many general knowledge graph databases and domain-specific knowledge bases have been released to facilitate research. We introduce more general and domain-specific knowledge bases in Appendices F-A1 and F-A2.

2.2 Definition and Notation

Knowledge graph reasoning refers to the process of using a knowledge graph, a structured representation of knowledge, as the basis for making logical inferences and drawing conclusions. More formally, the research question can be defined as

DEFINITION 1. (Knowledge Graph) Let $G = (V, E, R)$ be a knowledge graph, where V is the set of entities, E is the set of relationships, R is the set of triples (v_i, r_j, v_k) denoting relationships between entities, where $v_i, v_k \in V$ and $r_j \in E$. **Knowledge graph reasoning:** Answer queries by traversing and reasoning over the graph.

DEFINITION 2. (Knowledge Graph Reasoning) Let $G = (V, E, R)$ be a knowledge graph, where V is the set of entities, E is the set of relationships, R is the set of triples (v_i, r_j, v_k) denoting relationships between entities, where $v_i, v_k \in V$ and $r_j \in E$. **Knowledge graph reasoning:** Answer queries by traversing and reasoning over the graph.

2.3 Symbolic Reasoning

Symbolic reasoning in knowledge graphs refers to the process of deriving logical conclusions and making inferences based on symbolic representations of entities, relationships, and rules within the graph structure. In this context, symbols represent entities or concepts, while relationships denote connections or associations between them. Symbolic reasoning involves applying logical rules and operations to manipulate these symbols, enabling the system to perform tasks such as deductive reasoning, semantic inference, and knowledge integration. By leveraging symbolic representations and logical reasoning, knowledge graphs can facilitate complex problem-solving, semantic understanding, and decision-making in various domains, ranging from natural language processing to artificial intelligence applications.

2.4 Neural Reasoning

Neural reasoning in knowledge graphs refers to the utilization of neural network models to perform reasoning tasks directly on the graph structure. Unlike traditional symbolic reasoning approaches, which rely on explicit rules and logical operations, neural reasoning leverages the power of deep learning techniques to learn implicit patterns and relationships within the graph. Through the use of neural networks, knowledge graphs can capture complex, non-linear dependencies between entities and infer higher-level knowledge from the graph's interconnected nodes. Neural reasoning methods often involve embedding entities and relationships into continuous vector spaces, allowing neural networks to efficiently process and reason over large-scale knowledge graphs. This approach enables knowledge graphs to handle uncertainty, noise, and incompleteness

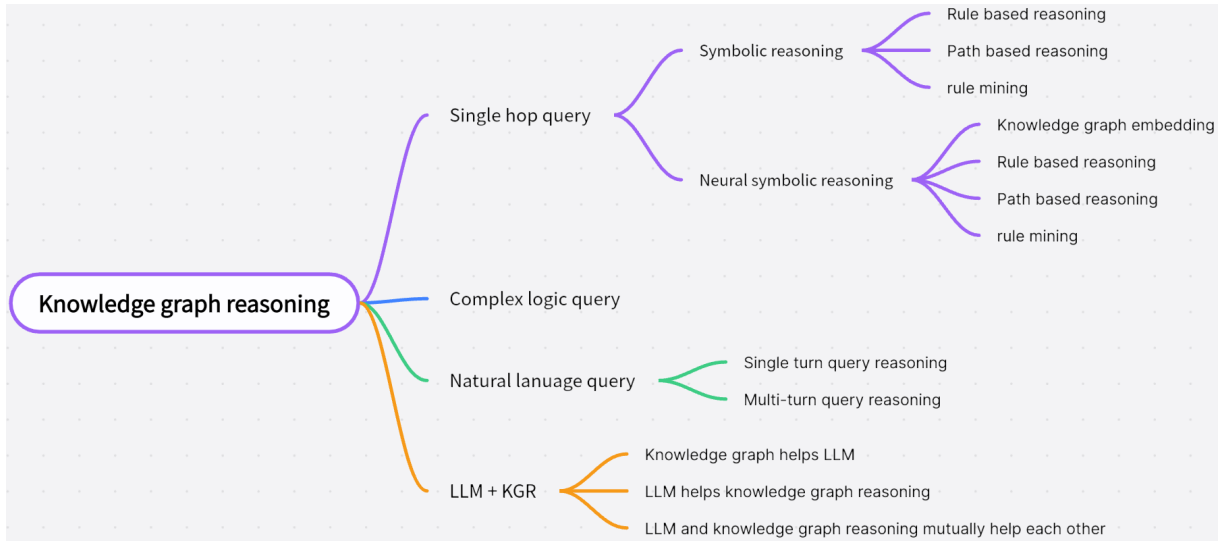


Figure 1: Survey framework.

in the data, making neural reasoning a promising paradigm for various applications, including question answering, recommendation systems, and semantic understanding.

2.5 Neural Symbolic Reasoning

Neural symbolic reasoning represents a fusion of neural network-based approaches with symbolic reasoning techniques, aiming to leverage the strengths of both paradigms in handling complex reasoning tasks. In this framework, neural networks are used to learn representations of symbolic entities and relationships within a knowledge graph, capturing both their semantic meanings and structural dependencies. These learned representations are then combined with symbolic reasoning mechanisms to perform logical inference and reasoning tasks. By integrating neural and symbolic components, neural symbolic reasoning approaches strive to overcome the limitations of each individual approach. Neural networks offer the ability to learn from data and handle uncertainty, while symbolic reasoning provides formal logic-based reasoning and interpretability. This hybrid approach has shown promise in various domains, including natural language understanding, knowledge graph reasoning, and automated theorem proving, by enabling more robust and flexible reasoning capabilities.

2.6 Deductive Reasoning

Knowledge graph deductive reasoning is a method used to derive new information from existing data within a knowledge graph by applying logical rules. A knowledge graph structures information as a network of entities and their interrelationships, represented in triples of subject-predicate-object. Deductive reasoning in this context involves using established logical rules to infer new facts. For instance, if the knowledge graph contains the triples "Alice works at XYZ Corp" and "XYZ Corp is located in New York," a rule might deduce that "Alice works in New York." This process leverages the structured nature of the graph and the logical relationships

between entities to expand the knowledge base, ensuring that new conclusions are logically consistent with the existing data.

2.7 Inductive Reasoning

Knowledge graph inductive reasoning involves deriving generalized conclusions from specific instances within a knowledge graph. Unlike deductive reasoning, which applies general rules to specific cases, inductive reasoning identifies patterns and regularities in the data to formulate broader generalizations or hypotheses. For example, if a knowledge graph contains numerous instances where employees of various companies in New York tend to have a high turnover rate, inductive reasoning might lead to the hypothesis that companies in New York generally experience higher employee turnover. This approach allows for the generation of new insights and predictive models by examining trends and correlations in the data, providing a foundation for further exploration and hypothesis testing within the structured framework of a knowledge graph.

2.8 Abductive Reasoning

Knowledge graph abductive reasoning is a process used to infer the most likely explanation for a given set of observations within a knowledge graph. This type of reasoning seeks to find the best hypothesis that explains the observed data, often dealing with incomplete or uncertain information. For example, if a knowledge graph shows that a person has visited multiple cities known for tech conferences, abductive reasoning might infer that the person is likely involved in the tech industry. Unlike deductive reasoning, which guarantees the truth of the conclusion if the premises are true, or inductive reasoning, which generalizes from specific instances, abductive reasoning focuses on finding the most plausible explanation. This method is particularly useful in situations where there are multiple possible explanations, and it aims to identify the one that best fits the available evidence within the structured relationships of a knowledge graph.